CODE :-

```
import numpy as np

patterns = [
    np.array([1, 1, 0, 0]),
    np.array([1, 1, 0, 1]),
    np.array([0, 0, 1, 1]),
    np.array([0, 0, 1, 0])
]

vigilance = 0.6
w = []

def match(x, w):
    y = np.dot(x, w)
    return np.sum(y) / np.sum(x)

cluster = []
for p in patterns:
    f = 0
    if len(w) == 0:
        print("match: 1 (first pattern with itself)")
    for i in range(len(w)):
        m = match(p, w[i])
        print("------------------------------------")
        print("match:", m)
        if m >= vigilance:
            print(p.tolist(), "-> same cluster", i)
            w[i] = np.minimum(w[i], p)
            print("w =", w[i].tolist())
            cluster[i].append(p)
            f = 1
            break
    if f == 0:
        w.append(p)
        cluster.append([p])
        print("w =", [arr.tolist() for arr in w])
        print(p.tolist(), "-> new cluster", len(w)-1)


print("****************************")
print("\nFinal clusters:")
for i in range(len(cluster)):
    print("------------------------------------")
    print("Cluster", i, ":")
    for p in cluster[i]:
        print(p.tolist())
```

OUTPUT :-

```
match: 1 (first pattern with itself)
w = [[1, 1, 0, 0]]
[1, 1, 0, 0] -> new cluster 0
-------------------------------------
match: 0.6666666666666666
[1, 1, 0, 1] -> same cluster 0
w = [1, 1, 0, 0]
-------------------------------------
match: 0.0
w = [[1, 1, 0, 0], [0, 0, 1, 1]]
[0, 0, 1, 1] -> new cluster 1
-------------------------------------
match: 0.0
-------------------------------------
match: 1.0
[0, 0, 1, 0] -> same cluster 1
w = [0, 0, 1, 0]

***************************
Final clusters:
-------------------------------------
Cluster 0 :
[1, 1, 0, 0]
[1, 1, 0, 1]
-------------------------------------
Cluster 1 :
[0, 0, 1, 1]
[0, 0, 1, 0]
```